

Running it *free*

The stack decision, the \$0 budget ledger with real numbers, the exact un-suspend runbook, and the guardrails that keep a free-tier trading system from suspending itself again.

Doc	Date	Target
05 of 06 — Infra & operations	2026-08-24	\$0/month · 2.3 GB bandwidth headroom

Keep the stack. Fix the behavior. Don't migrate.

Everything already runs on services with workable free tiers, hand-configured secrets, and working auth. A migration buys new failure modes, not savings — the outage was caused by **our streams**, not the host.

SERVICE	ROLE	WHY IT STAYS	WATCH-OUTS
Render (free web service + Key Value)	The one backend process: API + daemon + Rust engine child · Redis (bus, hot state, pulse plane)	Already configured by hand with 8 dashboard secrets incl. <code>VAULT_ENC_KEY</code> ; docker image builds Rust+Python; resume is one dashboard action	5 GB/mo bandwidth · 750 h/mo per workspace (CryptAI must be its only always-on service) · sleeps without keepalive · never recreate from Blueprint
Supabase (free)	Google OAuth + JWT · Postgres (sessions, proposals, trades, prefs)	Auth the product needs is already built against it; JWKS verification means the backend holds no service key	ACTIVE_HEALTHY (verified 2026-08-24) — the live frontend's auth traffic keeps it awake; watch the 7-day pause rule after any frontend outage · 500 MB DB · transaction pooler :6543 only
Vercel (Hobby)	Next.js frontend — already live	Zero-config deploys, already wired to Supabase + backend	Hobby is licensed non-commercial — fine for a free beta; budget Pro (\$20/mo) the day payments ship
OpenRouter (:free rotation)	All three LLM call sites	6-model free rotation wired with skip-on-dead + loud total-failure; \$0	Request caps are the real limit: ~50/day uncredited, ~1,000/day after a one-time \$10 credit — buy it
GitHub Actions	CI + the 10-min keepalive that keeps monitors alive	Public repo = free minutes	Scheduled workflows run from the default branch (still v2!) and auto-disable after 60 idle days
Binance (public data)	Market data — no key, no billing	Free; limits are IP rate weights, not money	Bytes are the budget: REST-poll everywhere except the viewer-gated dashboard feed

Contingency (only if Render bites twice): Oracle Cloud Always-Free ARM VM (4 OCPU/24 GB, 10 TB egress) can host the whole compose stack — real ops burden, keep it as plan B, not plan A. Dormant configs to delete: Pinecone keys (zero call sites). Optional later: Sentry free tier (SDK already ships).

The bandwidth ledger: ~2.8 GB against a 5 GB cap

Render's 5 GB meter counts **all** egress plus ingress on service-opened connections — service-initiated traffic is simply where ours went (5.61 of 5.63 GB on Aug 10). Red-team correction: the original ledger missed its biggest post-fix line — **LLM request bodies are egress too**.

MONTHLY PLAN OF RECORD (AFTER R0 + R1)

LLM egress — 40k-token prompts ≈ 160KB/call × ~560 calls/day (hot-window model)		~1.00 GB
Viewer-gated dashboard feed (~2 h/day viewing)		0.55 GB
Engine book poll (1 batched req / 5s — the staleness heartbeat)		0.50 GB
Engine funding/OI poll (2 req/symbol/60s)		0.40 GB
Position-mode price streams + klines by REST + Supabase pooler		~0.20 GB
JWKS (cached per process) + keepalive		0.02 GB

TOTAL ≈ 2.7 GB / mo

HEADROOM ≈ 2.3 GB (46%)

The LLM line is why the **prompt shrink moved from R2.8 into R1** (40k → ≤5k tokens cuts it ~8×) and why `ANALYSIS_KEEP_WARM` must stay **false** on free tier — keep-warm at 24/7 cadence is ~6.9 GB/mo of prompts alone: re-suspension by LLM egress. Unit economics: one always-on kline WS ≈ 0.6 GB/mo; depth20@100ms ≈ 0.7 GB/day/symbol — never always-on.

OTHER FREE-TIER METERS	
METER	POSITION
Render hours	~730/750 — CryptAI must be the workspace's only always-on service
Render KV	25 MB allkeys-lru, no persistence — eviction can hit <code>engine:on</code> (silent 503 outage), <code>cmd_reply</code> (approval 504s), desk state. R1.7 hardens: candle bus <code>persist=False</code> , <code>engine:on</code> re-asserted, watch <code>evicted_keys</code> ; Postgres is the only rehydration truth
Supabase DB	500 MB — years of headroom at this scale
Supabase MAU	50k — waitlist is 50
Vercel	100 GB/mo — ample
OpenRouter	~1,000 req/day with \$10 credit; hot-window clustering keeps 50 users ≈ 560/day

Bringing production back, in the only safe order

1

Land R0 first

Do not resume before the DataAgent stream fix merges. Resuming HEAD as-is re-suspends the workspace in ~2–3 days (autoDeploy is on — resume deploys HEAD immediately).



2

Env deltas first

On the still-suspended service:

`TRADING_SYMBOLS=BTCUSDT,ETHUSDT,SOLUSDT`
 · `DAEMON_DISABLE_AGENTS=memory` (512MB!) ·
 verify `ADMIN_USER_IDS` ·
`ANALYSIS_KEEP_WARM` stays false — keep-warm is ~6.9 GB/mo of LLM egress. Supabase is **ACTIVE_HEALTHY** (verified 2026-08-24) — just confirm, don't restore.



3

Resume — ONLY two things

Billing → settle the overage → **Resume ONLY cryptai-backend + cryptai-redis; klaro + SlackBot stay suspended** — three always-on services ≈ 2,232 h/mo vs the 750 h cap: a second suspension, on hours.



4

Schema truth

Prod tables exist only via boot-time `checkfirst` self-heal. Verify `sessions / proposals / pulse_snapshots` exist (information_schema or endpoints), then `alembic stamp head` once so future migrations run clean.



5

Verify + watch

`/health/ready` →
 redis+db true · session endpoints → 401 · CORS → 200 · "Daemon started" + monitors online · **exactly one** web service deployed · bandwidth graph ≈ flat over 48h · steady-state **RSS < ~400 MB** · Redis `evicted_keys = 0`.

⚠ NEVER RECREATE THE SERVICE

cryptai-backend was configured **by hand**, not from the Blueprint. Recreating mints a new URL and silently drops the 8 dashboard-only secrets — including `VAULT_ENC_KEY`, which encrypts every user's exchange keys. **Losing it orphans every stored credential, unrecoverably.** Resume; never delete-and-recreate.

ENABLING THE SIGNAL ENGINE (R1, SEPARATE STEP)

- Precondition: the Pulse-dataclass fix + klines-to-REST are merged.
- Set `RUN_SIGNAL_ENGINE_IN_API=true` + explicit `SYMBOLS` (binary defaults to 5 symbols — more bandwidth than planned).
- Watch logs for pulse publishes, **then** flip `SIGNAL_PLANE_ENABLED=true`. Never both blind — a dead plane silently drains user clocks (skipped_stale).

BRANCH & KEEPALIVE TRUTH

- GitHub default branch is **still v2** — scheduled workflows run from it, so the live keepalive is the old always-green version. Make the m1 line the default (R1) to activate the fail-loud keepalive.
- GitHub auto-disables cron workflows after 60 idle days — the keepalive is load-bearing (its death sleeps the daemon and kills monitors). Commit at least every 2 months or re-enable manually.
- Rotate the Render API key (it was pasted into a chat on 2026-08-03).

Secrets and config: what lives where, and what must never move

THE 8 DASHBOARD SECRETS (SYNC:FALSE – NAMES ONLY)		
NAME	ROLE · CRITICALITY	
VAULT_ENC_KEY	Fernet master key for user exchange credentials — loss is unrecoverable. Back it up to a password manager today; a key-rotation path is roadmap debt	
POSTGRES_URL	Supabase transaction pooler :6543 + statement_cache_size=0 — never revert either half alone	
SUPABASE_URL	JWKS verification base — backend holds no service key by design	
OPENROUTER_API_KEY	Without it, session start correctly 503s (capacity gate)	
API_AUTH_TOKEN	Service/admin token — admin-capable; slated for replacement (audit #6)	
ADMIN_USER_IDS	Fail-closed: empty = nobody is admin	
API_CORS_ORIGINS	Must include cryptai-app.vercel.app	
MULTI_USER_SEED_IDS	Optional demo accounts	

- CONFIG TRUTHS

 - render.yaml is documentation, not the control plane — the live service was hand-configured; dashboard edits change production, blueprint edits don't. Keep them in parity by review.
 - Frontend env: NEXT_PUBLIC_API_TOKEN must stay unset forever — anything NEXT_PUBLIC_* ships in public JS, and this one is admin-capable.
 - Money gates stay untouched: USE_TESTNET=true , LIVE_TRADING_CONFIRMED=false , ENABLE_EXECUTION unset — never flipped to make anything "work".
- OPS RHYTHM (10 MIN/WEEK)

 - Render bandwidth graph — the one chart that predicts suspension; alarm bell at >100 MB/day sustained
 - Keepalive workflow green + not auto-disabled
 - Supabase DB size + project not paused
 - OpenRouter request usage vs daily cap
 - After any deploy: /health/ready + monitors-online log line